Publication List

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Summary: 3 peer-reviewed articles, 4 under review (two single-author), 3 in preparation.
Research areas: parity-violating (Chern–Simons) modified gravity, black-hole perturbation theory and quasinormal modes, gravitational-wave phenomenology, QFT in curved and time-dependent backgrounds.

Peer-Reviewed Journal Articles

[1] **A. Rout** and B. Altschul, "First-Order Axial Perturbation of the Reissner–Nordström Metric Against a Possible Parity-Violating Gravity Background," *Symmetry* **18**, 560 (2026).
doi:10.3390/sym18040560 · arXiv:2510.01106 [gr-qc]

[2] **A. Rout** and B. Altschul, "Primordial Gravitational Wave Birefringence in a de Sitter Background with Chern–Simons Coupling," *Universe* **12**, 6 (2026).
doi:10.3390/universe12010006 · arXiv:2512.03919 [gr-qc]

[3] **A. Rout** and B. Altschul, "Bound States and Particle Production by Breather-Type Background Field Configurations," *Symmetry* **16**, 1571 (2024).
doi:10.3390/sym16121571 · arXiv:2410.09273 [hep-th]

Submitted Manuscripts

[4] **A. Rout**, SINGLE AUTHOR
"Analytic Parity-Violating Greybody Factors of a Dynamical Chern–Simons Black Hole," submitted to *General Relativity and Gravitation* (2026).
arXiv:2610.10280 [gr-qc]

[5] **A. Rout** and B. Altschul, "Gravitational Wave Parity-Violating Strain from Pulsar Glitches in Chern–Simons Modified Gravity," submitted to *Physical Review D* (2026).
arXiv:2608.24002 [gr-qc]

[6] **A. Rout**, SINGLE AUTHOR
"Isospectrality Breaking in the Axial Quasinormal Modes of Reissner–Nordström Black Holes in Dynamical Chern–Simons Gravity," submitted to *Physical Review D* (2026).
arXiv:2610.38742 [gr-qc]

[7] **A. Rout** and B. Altschul, "Mixed-Order Parity-Violating Corrections to Rotating Black Holes in Combined Scalar–Gauss–Bonnet and Dynamical Chern–Simons Gravity," submitted to *Physical Review D* (2026).
arXiv:2611.04945 [gr-qc]

In Preparation

[8] **A. Rout** and B. Altschul, "Rotation-Induced Isospectrality Breaking of Charged Black Holes in Dynamical Chern–Simons Gravity," in preparation.

[9] **A. Rout**, "Quasinormal-mode-driven pseudoscalar radiation in dynamical Chern–Simons ringdown," in preparation.

[10] **A. Rout** and B. Altschul, "Parity-violating spherical photon and timelike orbits in dynamical Chern–Simons gravity: a non-degenerate, sector-separating discriminator from linear to quadratic order in spin," in preparation.